Project: To-Do List API **CR ID:** CR-004 **Title:** Audit Trail for Task Changes **Type:** Enhancement — Traceability **Priority:** Medium **Status:** Proposed

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1. Problem Statement

The system does not keep any historical record of changes made to a task. Once a task's title, status, or due date is overwritten by an update, the previous value is gone — there is no way to answer "what did this task look like before, and when did it change?"

2. Proposed Change

Record a chronological history of create, update, and delete operations performed on tasks, capturing what changed, when, and (to the extent the system can identify it) by whom. This history is read-only to API consumers — it is a record of what happened, not something clients can edit directly.

3. Scenario

1. A task is created with `status = TODO`. An audit entry is recorded: operation `CREATE`, new values include `status = TODO`.
2. Later, its status is changed to `IN_PROGRESS` via `PUT`. A second audit entry is recorded: operation `UPDATE`, old value `status = TODO`, new value `status = IN_PROGRESS`.
3. Still later, its due date is changed. A third audit entry is recorded: operation `UPDATE`, old/new `dueDate` values.
4. A caller requests the task's audit history and receives all three entries in chronological order, each showing what changed and when.

4. Functional Requirements

- **FR-01:** The system shall record an audit entry whenever a task is created.
- **FR-02:** The system shall record an audit entry whenever a task is updated, capturing both the old and new values of the fields that changed.
- **FR-03:** The system shall record an audit entry whenever a task is deleted, capturing the task's state immediately before deletion.
- **FR-04:** Every audit entry shall record the affected task's identifier, the operation type, a timestamp, and an actor identity.
- **FR-05:** Audit entries shall be persisted durably (Oracle) and shall survive the task they describe being deleted — a deleted task's audit history is not removed along with it.
- **FR-06:** Audit entries shall be written as part of the same transaction as the operation they describe — if the underlying create/update/delete fails or rolls back, no audit entry for it shall be persisted.
- **FR-07:** A read-only endpoint shall allow retrieving a task's audit history in chronological order.
- **FR-08:** Audit entries shall not be directly creatable, modifiable, or deletable through any API exposed to clients.

5. Acceptance Criteria

- Creating, updating, and deleting a task each produce exactly one corresponding audit entry with correct old/new values.
- A task's full audit history can be retrieved and is ordered chronologically.
- Deleting a task does not delete its prior audit entries.
- No API endpoint allows writing directly to the audit history.
- A failed/rolled-back task operation produces no orphaned audit entry.

6. Scope

In Scope:

- Auditing `Task` create, update, and delete operations.
- A read-only task-level audit history endpoint.

Out of Scope:

- Auditing `Category` operations (not requested; can be a future, structurally similar extension).
- A UI for browsing audit history.
- Per-field-level granular permissions on who can view audit history (no authentication system currently exists in this project — see `TECHNICAL_DESIGN.md` for how actor identity is handled in its absence).

7. Approval

- **Business Approval:** Pending
- **Technical Approval:** Pending
- **Implementation Status:** Not Started